IS THERE A "LANGUAGE OF MUSIC-VIDEO CLIPS" ? A QUALITATIVE AND QUANTITATIVE STUDY

Laure Prétet
LTCI, Télécom Paris
Bridge.audio, Paris, France

Gaël Richard
LTCI, Télécom Paris
IP Paris, France

Geoffroy Peeters
LTCI, Télécom Paris
IP Paris, France

ABSTRACT

Recommending automatically a video given a music or a music given a video has become an important asset for the audiovisual industry - with user-generated or professional content. While both music and video have specific temporal organizations, most current works do not consider those and only focus on globally recommending a media. As a first step toward the improvement of these recommendation systems, we study in this paper the relationship between music and video temporal organization. We do this for the case of official music videos, with a quantitative and a qualitative approach. Our assumption is that the movement in the music are correlated to the ones in the video. To validate this, we first interview a set of internationally recognized music video experts. We then perform a large-scale analysis of official music-video clips (which we manually annotated into video genres) using MIR description tools (downbeats and functional segments estimation) and Computer Vision tools (shot detection). Our study confirms that a "language of music-video clips" exists; i.e. editors favor the co-occurrence of music and video events using strategies such as anticipation. It also highlights that the amount of co-occurrence depends on the music and video genres.

1. INTRODUCTION

Each day, an ever-growing quantity of videos is created by professionals (for advertisement, movies, series, etc) and individuals (for Instagram, TikTok, YouTube, etc). Finding an appropriate soundtrack to emphasize the video content is therefore a common exercise, which can be time-consuming if done manually. This explains the success of commercial systems such as MatchTune or of research papers such as “Look, Listen and Learn” [1]. While such systems are very good at recommending music based on the video content, the temporal synchronization between both modalities is rarely taken into account. In order to develop synchronization-aware recommendation systems, some domain knowledge is required on how the synchronization is performed in real videos that feature music. In this work, we attempt at bridging this knowledge gap by performing a fine-grained cross-modal analysis of the synchronization between audio and video content. We hypothesize that better understanding professionally produced music videos helps designing better models for music-video synchronization. This has applications in automatic music-video recommendation [2–6] and generation [7–9].

Temporal structure (at the beat, bar or functional segment level) is one of the dominant characteristics of music. For this reason, its automatic estimation has received a lot of attention in the Music Information Retrieval (MIR) community [10]. Temporal structure in video (cuts, scenes, chapters) has similarly received a lot of attention in the Computer Vision community (for example with the goal of creating video summary [11]). Our fine-grained analysis will be using these structural elements.

Our cross-modal analysis could be performed on any type of video that features a musical soundtrack (eg commercials, movies). We focus here on the special case of of Official Music Videos (OMV). We call OMV an audiovisual document where the audio part consists in a music track, and which aims at promoting said track and its performing artists. As a result, the music track is generally the only source of audio in OMVs. This makes OMVs good prototypes for a study on music-video synchronisation. We do not consider user-generated videos, because we assume that analyzing professionally produced OMVs is more likely to provide reusable insights.

In the specific case of OMVs, the editing team will often arrange the video rushes based on the structure of the music track [12]. In some cases, the music track can also be adapted from the studio version for narrative purposes. Therefore, music and video structure are de facto associated. However, the level of synchronicity is not always the same, depending on the considered OMV. This is not only due to artistic choices but also depends on the music genre and video genre, as we will see in our study.

Proposal and paper organization. In this paper, we study the relationship between music and video temporal organization using a qualitative and a quantitative approach. The *qualitative* study is based on a set of interviews with three renowned specialists of official music videos. We interview them in order to find out if and how they consider the relationship between music and video structure in their work. The *quantitative* analysis is based on a detailed analysis of music and video structural events in OMV using MIR and Computer Vision

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tools. The OMVs correspond to a subset of the Harmonix dataset [13]. We study specifically the relationship between the duration of music and video segments and between the positions of their respective boundaries. We highlight the dependency of those according to the OMV music and video genre (for which we annotated the data).

The paper is organized as follows. Section 2 discusses the related literature. Section 3 describes the qualitative study and summarizes the interviews of three music video experts: Jack Bartman (composer), Alexandre Courtès (director) and Maxime Pozzi (editor). Section 4 describes the quantitative study: the dataset creation (4.2), the analysis of the music and video segment duration (4.3.1) and of the music and video segment position (4.3.2). Section 5 concludes and discusses the perspectives of this work.

2. RELATED WORK

2.1 Music-Video Synchronization: A Short Review

Music supervision is an industry that aims specifically at synchronizing music to video. Music supervision experts are dedicated to proposing the best soundtrack to all types of videos, ranging from commercials to movies and series. As of today, this recommendation and synchronization work still features a large amount of manual work. Inskip et. al. [14] interviewed music supervision experts and described their workflow. The authors mention that "the clearest briefs appear to be moving images", suggesting that other types of data (emotion, textual description, reference soundtracks) are not necessary to perform the task.

At the same period, Gillet et al. [15] proposed a system that can automate part of the music supervision task. Their system relies on the synchronization of music structure (onsets and functional segments) and video structure (motion intensity and cuts) to perform music-video recommendation, without external data. Yang [16] and Mulhem [17] proposed similar approaches.

More recently, Alexander Schindler gathered a dataset of OMVs (the Music Video Dataset) and performed an in-depth analysis of this specific media [18]. In [19], Schindler and Rauber explain how shot boundary detection is an ill-defined task in music videos, as shot transitions are used in a complex and artistic way. By analyzing the clips, they observe that the music videos present characteristic editing styles (number of shots per second, types of transition) for certain music genres or moods. But they do not quantify this correlation. In [12], the same authors analyze the correlation between visual contents (objects present in the scene) and music genre. For example, cowboy hats are almost systematic in country music videos.

In our study, we propose a joint approach. We analyze the correlation between the music/video *structure* and music/video *genres*.

2.2 Audiovisual Structure Estimation Tools

Our quantitative study (Section 4) relies both on MIR to analyze the music structure and on Computer Vision to analyze the video structure. More specifically, we estimate the downbeat positions, functional segments and shot boundaries from the OMV of our dataset. In the following, we describe the tools we have used for our analysis.

Downbeat tracking is a popular MIR task [20]. As a result, several ready-to-use libraries are available to estimate downbeat positions from audio files [21, 22]. The state-of-the-art algorithm of Böck et al. [23] consists in two steps. First, a `RNNDownBeatProcessor`, which relies on multiple Recurrent Neural Networks (LSTMs), estimates jointly beat and downbeat activation functions. The output of the neural networks represents the probability of each frame of being a beat or downbeat position. These activation functions are then fed as observations to a `DBNDownBeatTrackingProcessor`, which relies on a Dynamic Bayesian network (DBN). The DBN outputs the beat positions of highest likelihood, along with their position inside the bar.

At a larger timescale, the automatic detection of boundaries between functional segments (choruses, verses and bridges) has also received a lot of attention from the MIR community. The Ordinal Linear Discriminant Analysis (OLDA) algorithm by McFee et al. [24] relies on supervised learning to perform this task. This method adapts the linear discriminant analysis projection by only attempting to separate adjacent segments. Then, the obtained features are clustered with a temporal constraint: only similar successive segments are merged together.

Similar to music, videos can be divided into segments of various duration, from shots to scenes to chapters and longer sequences. In this study, we focus on a segmentation into shots. The TransNet system [25], by Souček et al., is a Convolutional Neural Network which employs dilated 3D convolutions and which is trained in a supervised way on a shot boundary detection task.

3. QUALITATIVE ANALYSIS: INTERVIEWS

3.1 Methodology

In order to gather intuition on the synchronization of music and video, we conducted a series of semi-structured face-to-face interviews. We selected three music video experts from different professions: composition, direction and editing. Following Inskip et. al. [14], we selected the respondents using a snowball sampling technique.

Interviews were performed using the Zoom video conferencing software, lasting up to one and a half hours. The interviews were transcribed manually by the researcher, and transcripts were sent back to the respondents for validation. Areas of discussion included the participant's day-to-day workflow and technical tools, their interactions with the other professions of the industry, and their opinion on example music videos prepared by the researcher.

3.2 Interviews Summary

3.2.1 Jack Bartman, Composer

As a composer (for commercials such as Nike, Apple or UbiSoft), Bartman has to adopt both a global and a pre-

cise local approach: the content of the music has to match the visual atmosphere, and its temporal structure must be aligned both globally at clip level and locally at frame level. In some cases, the editing follows the structure of the music. But in other cases, typically for advertisement, it is the opposite, and the composer has to adapt the music to an existing movie. Most of the time, when music has to be edited on an existing movie, the slicing operation is privileged.

"Slicing can happen at unconventional moments, like the first or last beat of a bar! I simply add a sound effect to make it work."

Time stretching and accelerations can be employed too, but are far less usual. Bartman stresses that synchronizing cuts to audio events is especially important around emotional climaxes of the video. Finally, for some projects, an exact synchronization is not the golden rule:

"This year, I worked on a short movie about psychological aspects of the Covid-19 lockdown. After getting used to an imperfectly synchronized mockup soundtrack, the director did not want to use the final version, as the mockup would better suit the intended "madness" atmosphere".

3.2.2 Alexandre Courtès, Director

As a director (such as for U2, Phoenix, Cassius, Franz Ferdinand or Jamiroquai), Courtès generally has a lot of freedom when it comes to the temporal organization of a music video. Directors often come up with their own concept and they have little constraint about the content of the video. At large temporal scale, their mission is to emphasize the music climaxes by the appropriate video content.

"The music video will often show a performance, so it is similar to a musical comedy: it has to feature costumes, chapters, sets, acts."

Directors are not responsible for placing the cuts, but they can introduce diversity in the video transitions (explosions, large objects passing in front of the camera; see [19] for a more exhaustive list).

"Cuts have to follow the music's rhythm, even though they might not always co-occur with beats."

3.2.3 Maxime Pozzi, Editor

As an editor (such as for Rihanna, Taylor Swift, Foals or Woodkid), Pozzi has to combine both a local, frame-level approach to the design of a global emotional trajectory.

"Editors and musicians have a similar job, we all want the same thing: rhythm, narration, climaxes."

For chorus and verses, the editing will follow the rhythm and typically accelerate near climaxes. During bridges, it will often be slower and poetic. This can be illustrated for example by Katy Perry's *Firework* music video (Figure 1). In this clip, we can see some functional segments where cuts happen very frequently (several times in each bar) and segments where they happen less frequently, for example on the downbeats only.

Editing can be used as an element of narration. For example, in Adele's *Rolling in the deep* music video, starting at timestamp 02:20, the cuts are systematically placed just before the downbeat (see Figure 2).

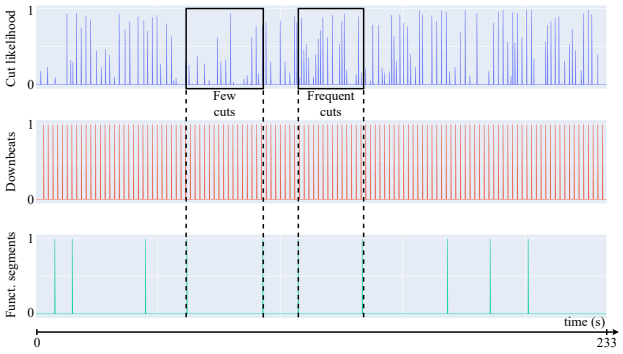


Figure 1: Audiovisual structure of Katy Perry, *Firework*, full clip. Horizontal axis: time. Cuts: TransNet estimates. Downbeats: Madmom estimates. Music functional segments: OLDA estimates.

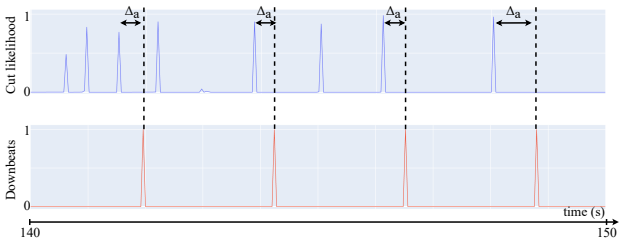


Figure 2: Audiovisual structure of Adele, *Rolling in the deep*, timestamps 02:20 to 02:30. Horizontal axis: time. Cuts: TransNet estimates. Downbeats: Madmom estimates. Δ_a: anticipation of cuts with respect to downbeat.

"Off-beat cuts are used to create dynamics: to surprise the viewer, and illustrate the music's emotional climax. It makes the video direction appear more "indie" as well, this can be required by the performing artists."

3.3 Summary

These three interviews provide us with a series of intuitions and hypotheses about the way audio and video are synchronized in music videos. First, musical structure such as chorus and verses are taken into account when directing a music video. Second, audio events such as rhythm, beat and downbeat are taken into account when editing a music video. Finally, according to the desired atmosphere, the audio and video structural events can be more or less perfectly synchronized.

4. QUANTITATIVE ANALYSIS

4.1 Methodology

In the following, we conduct a set of quantitative experiments on how the Structural Events (SE) of the music and of the video are synchronized in time. We do so using Official Music Videos (OMVs). We therefore first collect a dataset of OMVs, along with music and video genre annotations (Section 4.2). For each of them we use MIR tools to estimate music SE (downbeats and functional segments) and Computer Vision tools to estimate video SE (shot boundaries). In our first experiment, we study the

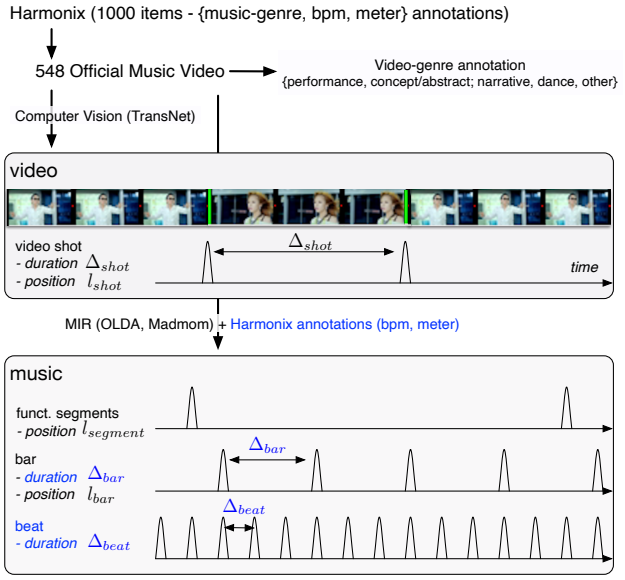


Figure 3: Schematic view of the different audiovisual structural events considered: shots $(\Delta_{shot}, l_{shot})$, functional music segments $(l_{segment})$, bars/downbeats (Δ_{bar}, l_{bar}) and beats $(\Delta_{beat}, l_{beat})$. Illustration music video: Psy, *Gangnam Style*.

correlation between the duration of the shots and the various musical SEs (beat and bar duration). In our second experiment, we study the temporal co-occurrence of the shot boundaries and the various musical SEs (bar and functional segment boundaries). We analyze the results of those for each music genre and each video genre.

4.2 Dataset

For our quantitative study, we consider a subset of the Harmonix dataset [13]. Harmonix was initially released for automatic estimation of beat, downbeat and functional music segments. It features popular (mostly hits) Western music tracks for which there is a high probability of having an associated music video. From the list of 1,000 YouTube video links provided, 899 were successfully retrieved, of which 40% contained only still images and 2.4% were duplicates. As a contribution of this work we provide the list and URLs of the remaining 548 OMVs as well as the genre annotations described below [1].

4.2.1 Annotations into Structural Events

We consider here two types of Structural Events (SE): those based on the music content -audio-, and those based on video content -image frames over time (see Figure 3).

Music SE. We consider three types of music SEs. At the smallest temporal scale we consider the beats and downbeats; at the largest temporal scale we consider the functional music segment boundaries (between the verses, bridges, choruses). Harmonix features a set of manual annotations [2]. However, these annotations correspond to studio versions of the tracks which can, in some cases, be largely different from the version used in the OMV. For this reason, we only used the annotations into bpm and meter of the Harmonix dataset to get the beat duration $\Delta_{beat} = \frac{60}{bpm}$ and bar duration $\Delta_{bar} = 4$ or $3\Delta_{beat}$ (which is computed as a multiple of the bar duration using the time signature). For the downbeat positions, we used the algorithm of Böck et al. [23], implemented in the Madmom library [21]. In the following, we denote by l_{bar} the list of downbeat positions for a given track. For the functional music segments, we used the implementation of OLDA from the MSAF library [26]. In the following, we denote by $l_{segment}$ the list of boundary positions between the segments for a given track. For our dataset, the average duration of functional music segments is 19.73 s. and the average bar duration is 2.30 s.

Video SE. We consider only the least ambiguous video SE, the shot boundaries (or cuts). To detect boundaries between shots, we use the TransNet system [25] and the associated library, available on GitHub [3]. The TransNet output is a continuous function of time $f_{shot}(t) \in [0, 1]$ representing the likelihood of a boundary at time t. f_{shot} has a sampling rate of 25 Hz.

Also, for each OMV, we compute the histogram of its shot duration. We do so by first estimating the list of shot boundary positions l_{shot} by thresholding $f_{shot}(t)$ with $\tau = 0.5$. The resulting segments have an average duration Δ_{shot} of 4.76s. We then compute the histogram of these durations. We denote by $\Delta_{shot}^{\max}$ the position of the maximum of this histogram (in seconds).

We sum up the various SE in Table 1.

Table 1: Notation associated to each SE considered.

Music		
genre		Harmonix annotations
funct. segments positions	$l_{segment}$	OLDA/MSAF
bar duration	Δ_{bar}	Harmonix annotations
bar/downbeat positions	l_{bar}	Madmom
beat duration	Δ_{beat}	Harmonix annotations
Video		
genre		Manual annotations
shot boundary probability	$f_{shot}(t)$	TransNet
shot boundary positions	l_{shot}	
most common shot duration	$\Delta_{shot}^{\max}$	

4.2.2 Annotations into genre

We consider both the genre associated to the music and the one associated to the video.

Music genre. While still controversial in its exact definition [27], music genre is a convenient way to describe musical content. For this reason, it has been and it is still a widely studied topic [4]. For our experiments, we use the music genre annotations provided by the Harmonix dataset metadata.

Video genre. Video genre classification is a much less studied topic. Existing studies focus on a much smaller sets of video genres [32–34]. Only Gillet et al. [2] and

[1] Our list is accessible at: https://gitlab.com/creaminal/publications/ismir-2021-language-of-clips/-/blob/master/video_genres.csv.

[2] into functional segments, downbeat and beat.

[3] https://github.com/soCzech/TransNet

[4] It has dedicated challenges [28], and large datasets featuring hundreds of categories [29–31].

Schindler [18] studied the case of OMVs and there is no consensus on their taxonomy of video genres. There is also no annotated dataset for this task.

We merge [2] and [18] to obtain a set of 5 video categories and and a corresponding single-label dataset. Maxime Pozzi, a professional music video editor, validated our taxonomy during our preliminary interview (see part 3.2.3). One author then manually annotated all 548 video clips of Harmonix into the five following video genres:

- `Performance videos (P)` : The artist or band are presented performing the song. 74 videos; example: Iron Maiden, *Powerslave*.
- `Concept/Abstract videos (C)` : The video illustrates the music metaphorically via a series of abstract shots related to semantics or atmosphere of the song. 227 videos; example: Lady Gaga, *Poker Face*.
- `Narrative videos (N)` : The music video has a strong narrative content, with identifiable characters and an explicit chronology. 160 videos; example: Taylor Swift, *Teardrops on My Guitar*.
- `Dance videos (D)` : Artists present a rehearsed dance choreography in sync with the music. 62 videos; examples: Sean Paul, *Get Busy*.
- `Other (O)` : Other types of music videos, including lyrics videos, animated music videos, etc. 25 videos; example: Train, *Hey, Soul Sister*.

4.3 Experiments

We hypothesize that the music structural events play an important role for the placement of cuts during the video editing. We check this assumption by measuring:

- if their segment duration are correlated in Section 4.3.1;
- if their position co-occur in Section 4.3.2.

According to Gillet [2], the performance of alignment-based music-video recommendation systems are strongly correlated to the video genre. We therefore differentiate our results by music and video genre.

4.3.1 Comparison between events duration

Our first experiment aims at evaluating to which extent the musical and video events have similar durations.

To measure this, we compare $\Delta_{shot}^{\max}$ (the most common shot duration) with the beat duration Δ_{beat} and bar duration Δ_{bar} obtained from the Harmonix annotations. When $\Delta_{shot}^{\max}$ is close to Δ_{bar}, this indicates that a systematic change of shots occurs with the same speed as the bar changes. This however does not mean that the changes occur simultaneously (we study this in Section 4.3.2).

This is for example the case of "Heartless" by Kanye West (see Figure 4 [top]) where the large peak at $\Delta_{shot}^{\max}$=2.72 s can be explained by the tempo at 88 bpm; or "Firework" by Katy Perry (see Figure 4 [bottom]) where the large peak at $\Delta_{shot}^{\max}$= 1.93 s can be explained by the tempo at 124 bpm.

In our dataset, a synchronization at the bar level ($0.5\Delta_{bar} < \Delta_{shot}^{\max} < 1.5\Delta_{bar}$) occurs for one fifth of the clips (95 music videos). Synchronization may also occur at other levels: at the beat level Δ_{beat}, or the pattern level $\Delta_{pattern}$ (usually an even multiple of the bar duration). In our dataset, a synchronization at the beat level ($0.5\Delta_{beat} < \Delta_{shot}^{\max} < 1.5\Delta_{beat}$) occurs for two thirds of the clips (329 music videos). However, synchronization at pattern level $\Delta_{pattern} = 4\Delta_{bar}$ almost never occurs (2 music videos).

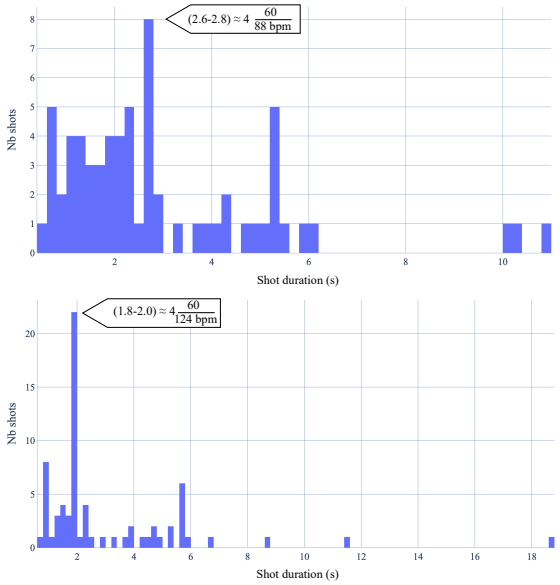


Figure 4: [top] Histogram of shot duration in the music video of *Heartless* by Kanye West. The tempo is 88 bpm. [bottom] Histogram of shot duration in the music video of *Firework* by Katy Perry. The tempo is 124 bpm.

In Table 2, we indicate for each music genre and video genre, the number of tracks for which the $\Delta_{shot}^{\max}$ correspond to Δ_{bar} or Δ_{beat}. We only focus here on the most represented genres, i.e. which appear at least 10 times. We observe a strong correspondence between $\Delta_{shot}^{\max}$ and Δ_{bar} for the music genres `Country`, `Dance/Electro` and `Rock` (one fourth of the tracks). We observe a strong correspondence between $\Delta_{shot}^{\max}$ and Δ_{beat} for the music genres `Alternative` and `Reggaeton` (three quarters of the tracks). This may imply, for example, that music video professionals favor more dynamic editing styles (using shorter shots on average) for `Reggaton` than for `Country` music. We observe a strong correspondence between $\Delta_{shot}^{\max}$ and Δ_{bar} for the video genre `Performance` (one fourth of the tracks). On the contrary, we observe a low correspondence between $\Delta_{shot}^{\max}$ and Δ_{beat} for the video genre `Other` (one third of the tracks). It is likely that music videos in the `Other` category favor experimental editing styles, with shots of more diverse duration.

As we see, there is a strong relationship between the video events and musical events duration. This however does not mean that the changes occur simultaneously. We study this in the next section.

4.3.2 Comparison between events position

Our second experiment aims at evaluating to what extent the musical events l_{seg}, l_{bar} and video events $f_{shot}(t)$ happen simultaneously. To measure this, we compute for each audio boundary i ($t_i \in l_{seg}$ or $t_i \in l_{bar}$) a score $S_i \in [0, 1]$.

Table 2: Agreement of musical structure (bar Δ_{bar} and beat Δ_{beat} level) and dominant shot duration $\Delta_{shot}^{\max}$ according to the **music genre** [top table] and according to the **video genre** [bottom table]. Highest values are highlighted in bold, lowest values in italic.

Music Genre	$\Delta_{shot}^{\max} \simeq \Delta_{bar}$		$\Delta_{shot}^{\max} \simeq \Delta_{beat}$	
	# tracks	%	# tracks	%
Alternative	2	8.3	19	**79.2**
Country	10	**29.4**	16	47.1
Dance/Electro	12	**24.5**	28	57.1
Hip-Hop	12	12.6	69	72.6
Pop	40	14.8	158	*58.3*
R&B	1	*5.3*	13	68.4
Reggaeton	1	8.3	9	**75.0**
Rock	4	**23.5**	10	*58.8*

Video Genre	$\Delta_{shot}^{\max} \simeq \Delta_{bar}$		$\Delta_{shot}^{\max} \simeq \Delta_{beat}$	
	# tracks	%	# tracks	%
Concept	33	14.5	148	**65.2**
Dance	11	17.7	40	**64.5**
Narration	28	17.5	93	58.1
Performance	19	**25.7**	41	55.4
Other	4	*16.0*	8	*32.0*

S_i is defined as the integral over time of the shot boundary likelihood $f_{shot}(t)$ tampered by a non-normalized Gaussian window $w(t)$. $w(t)$ is centered on t_i, with $\sigma = 2$ (such that the effective duration of the window is approximately 0.5s at a frame rate of 25Hz) and with $w(0) = 1$.

$$S_i = \int_t w(t - t_i) f_{shot}(t) dt, \quad \forall t_i \in \{l_{seg}, l_{bar}\}$$

A large value of S_i indicates that the t_i position (the music structural event) corresponds to a large probability of shot boundary. We then average S_i for all audio boundaries i to get S. S might be considered as a measure of precision, since it provides information on *how many audio boundaries are explained by a video boundary*. It should be noted that the number of video boundaries is larger than the number of audio boundaries (as seen in Figures 1 and 2). S is also close to the measure proposed by [35] to evaluate the performances of beat-tracking algorithms. A large value of S indicates that the shot boundaries are located at the same positions as the music structural events l_{seg} or l_{bar}. We compute S separately using the t_i from l_{seg} or from l_{bar}. To check if the amount of music-video event synchronization depends on the music and video genre, we average S over all tracks of a given genre (music or video).

Co-occurrence of music/video events by music genre. Table 3 [top part] shows the co-occurrence scores S aggregated over music genres. We observe variations of the values of S according to the music genre. For `Pop`, $S(l_{seg})$ is large (0.36) indicating that many shot transitions occur at the functional segment boundaries positions. For `R&B` and `Reggaeton`, $S(l_{bar})$ is large (0.31 and 0.28) indicating that many shot transitions occur at the downbeat positions. We also observe that the value of $S(l_{seg})$ and $S(l_{bar})$ vary according to the music genre with very small values for `Dance/Electronic`, `Hip-Hop` and `Rock`. This comes as a surprise especially for `Dance/Electronic`, because in the previous experiment, we observed a strong correspondence between the duration of shots and bars for this music genre. This shows that even though bars and shots have similar duration, their boundaries might not always co-occur.

Table 3: Shot transition intensity S around music boundaries (either functional segments boundaries l_{seg} or bar boundaries l_{bar}) according to **music genre** [top table] and according to the **video genre** [bottom table]. Mean values and confidence intervals at 95% are displayed. Highest values are highlighted in bold, lowest values in italic.

Music Genre	$S(l_{seg})$	$S(l_{bar})$	# tracks
Alternative	0.22 ± 0.08	0.23 ± 0.02	24
Country	0.20 ± 0.06	0.21 ± 0.02	34
Dance/Electro	*0.18 ± 0.05*	0.21 ± 0.02	49
Hip-Hop	0.19 ± 0.03	0.25 ± 0.01	95
Pop	**0.36 ± 0.02**	0.21 ± 0.01	271
R&B	**0.29 ± 0.10**	**0.31 ± 0.03**	19
Reggaeton	0.24 ± 0.11	**0.28 ± 0.04**	12
Rock	*0.18 ± 0.07*	*0.19 ± 0.03*	17

Video Genre	$S(l_{seg})$	$S(l_{bar})$	# tracks
Concept	0.20 ± 0.02	**0.23 ± 0.01**	227
Dance	**0.18 ± 0.04**	**0.24 ± 0.01**	62
Narration	**0.18 ± 0.03**	**0.23 ± 0.01**	160
Performance	*0.15 ± 0.04*	0.16 ± 0.01	74
Other	*0.11 ± 0.06*	*0.11 ± 0.02*	25

Co-occurrence of music/video events by video genre. Table 3 [bottom part] shows the co-occurrence scores S aggregated over video genres. We observe variations of the values of S according to the video genre. We see that the `Dance` video genre has a large value of $S(l_{bar})$ (0.24), which is not surprising given that video labeled as `Dance` actually show people dancing on the beat. We also observe large values of $S(l_{bar})$ for the `Concept` and `Narration` video genres with consistent synchronization on the downbeats. For the `Performance` video genre (the band is playing in front of the camera), we don't observe such a large correspondence ($S(l_{bar}) = 0.16$). For the `Other` video genre, the low values ($S(l_{bar}) = S(l_{seg}) = 0.11$) are not surprising, given that some videos are very experimental and may feature complex video transitions, which may be difficult to detect by the TransNet.

5. CONCLUSION

According to the professionals and to our experiments, official music videos are edited by taking into account the music structure. Although some experts mentioned that synchronization was often a matter of taste and intuition, we were able to bring out some trends. We showed that the co-occurrence of music and video structural events would vary according to the music and video genres. These elements can be reused to design or improve automatic music-video recommendation systems. For example, if the task is to recommend an illustration video for a Pop or R&B track, the system is expected to favor candidates that allow high synchronization of the structural events.

However, we have the intuition that other factors may impact the editing style of OMV. In future work, we plan to investigate the role of other metadata, such as release date, artist popularity or harmonic complexity. Although we focused on OMV for this study, we believe that a similar analysis can be conducted on other types of musical videos, e.g. movies or commercials.

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